A Project Report on

Chest X-ray Image Enhancement and Classification Using Convolution, Machine Learning, and Deep Learning in Python

Epbl-DataStructures using Python

Submitted by

V.Srinivas Raghavan	23881A04C4
K.Sonali	23881A04G5

Course Facilitator

Mr.K.Murali
Assistant Professor

Department of Electronics and Communication Engineering

Vardhaman College of Engineering, Hyderabad

March 2026

Contents

Abstract

Chest X-ray imaging is one of the most widely used diagnostic techniques for detecting lung diseases such as pneumonia, tuberculosis, and other respiratory disorders. The quality of X-ray images significantly influences the accuracy of diagnosis, making image enhancement and classification essential steps in medical image analysis. This project presents a Python-based approach for Chest X-ray Image Enhancement and Classification using convolution, machine learning, and deep learning concepts.

The proposed system begins by loading grayscale chest X-ray images and preprocessing them through Gaussian filtering to remove noise while preserving important anatomical structures. Edge detection is then performed using a custom convolution kernel implemented through both a brute-force convolution algorithm and an optimized OpenCV convolution function. The execution time of both methods is compared to demonstrate the efficiency of optimized image processing techniques.

After image enhancement, statistical image features, including mean intensity, standard deviation, and intensity range, are extracted to represent each image numerically. These features are normalized and used as inputs to a simple neural network–style logistic regression classifier implemented from scratch using NumPy, without relying on external machine learning libraries. The classifier is trained using gradient descent, and its performance is evaluated through loss and accuracy curves generated during training.

The project demonstrates the integration of digital image processing, convolution operations, feature extraction, and machine learning principles into a unified workflow for chest X-ray analysis. Although the current implementation uses sample labels for demonstration purposes, the framework can be extended to real medical datasets for disease classification. The project provides a practical understanding of image enhancement, feature engineering, and basic deep learning concepts while highlighting the role of artificial intelligence in medical image analysis.

Keywords: Chest X-ray, Medical Image Processing, Image Enhancement, Gaussian Filtering, Convolution, Edge Detection, Feature Extraction, Machine Learning, Deep Learning, Logistic Regression, Image Classification.

1 Introduction

Chest X-ray imaging is one of the most widely used medical imaging techniques for diagnosing respiratory diseases such as pneumonia, tuberculosis, lung infections, and other pulmonary disorders. It is a fast, non-invasive, and cost-effective diagnostic tool that helps healthcare professionals detect abnormalities in the lungs and chest cavity. However, the quality of chest X-ray images may be affected by noise, low contrast, and imaging artifacts, making accurate diagnosis more challenging. Image enhancement techniques, including Gaussian filtering and convolution-based edge detection, play an important role in improving image quality by reducing noise and highlighting significant anatomical structures. These preprocessing techniques provide clearer images that support both manual diagnosis and automated medical image analysis.

With the rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), automated analysis of chest X-ray images has become increasingly effective in assisting disease detection and classification. This project presents a Python-based framework that combines image enhancement, convolution operations, feature extraction, and a logistic regression classifier implemented using NumPy. Gaussian filtering is used for noise removal, while both manual and optimized convolution methods are applied to compare computational performance and enhance image features. Statistical features extracted from the processed images are normalized and used for classification, with training performance evaluated through loss and accuracy curves. The proposed approach demonstrates how digital image processing and machine learning techniques can be integrated to develop an efficient and scalable chest X-ray image analysis system.

2 Literature Review

Medical image analysis has become an important area of research due to the increasing use of Artificial Intelligence (AI) in healthcare. Several studies have demonstrated that image preprocessing techniques such as Gaussian filtering, histogram equalization, and convolution-based edge detection significantly improve the quality of chest X-ray images by reducing noise and enhancing important anatomical features. These enhancement techniques help improve the accuracy of disease detection and provide better inputs for automated classification systems. Researchers have also explored various feature extraction methods, including statistical, texture, and shape-based features, to represent chest X-ray images effectively for machine learning algorithms.

Recent advancements in Machine Learning (ML) and Deep Learning (DL) have further improved the performance of automated chest X-ray classification systems. Traditional machine learning methods such as Logistic Regression, Support Vector Machines (SVM), and Decision Trees rely on handcrafted features for classification, while deep learning models, particularly Convolutional Neural Networks (CNNs), automatically learn discriminative features directly from medical images. The proposed framework demonstrates the fundamental concepts of medical image processing and machine learning while providing a foundation for future development using advanced deep learning techniques.

3 Objectives

- To develop a Python-based system for chest X-ray image enhancement and classification.
- To improve image quality using Gaussian filtering and convolution-based edge detection techniques.
- To extract meaningful statistical features from the enhanced chest X-ray images for classification.

- To implement a logistic regression classifier and evaluate its performance using accuracy and loss metrics.
- To compare the execution time and efficiency of manual convolution and optimized OpenCV convolution methods.

4 Methodology

4.1 Block Diagram

The system consists of:

- **Chest X-ray Dataset**:Contains chest X-ray images used as input for image enhancement and classification.
- **Image Acquisition**: Loads the images from the dataset and resizes them to a standard size for processing.
- **Preprocessing**: Applies Gaussian filtering to remove noise and improve image quality.
- **Convolution**: Performs edge detection using manual and optimized convolution methods to enhance important image features.
- **Feature Extraction**: Extracts statistical features such as mean intensity, standard deviation, and intensity range.
- **Feature Normalization**:Normalizes the extracted features to ensure efficient and stable model training.
- **Classification**:Uses a logistic regression model implemented with NumPy to classify the processed chest X-ray images.
- **Evaluation and Visualization**:Evaluates the model using accuracy and loss metrics and displays the processed images along with the training curves.

4.2 Working Principle

The proposed system begins by loading chest X-ray images from the dataset and resizing them to a standard resolution for uniform processing. The images are then preprocessed using Gaussian filtering to remove unwanted noise while preserving important anatomical structures. This preprocessing step improves image quality and provides a cleaner input for subsequent analysis. After noise removal, convolution-based edge detection is applied using both a manually implemented convolution algorithm and an optimized OpenCV convolution function. The convolution operation enhances the edges and important lung features, making abnormalities more distinguishable. The execution time of both convolution methods is also compared to evaluate computational efficiency.

After image enhancement, statistical features such as mean intensity, standard deviation, and intensity range are extracted from the processed images. These features are normalized to improve the stability and performance of the classification model. The normalized feature vectors are then used to train a logistic regression classifier implemented using NumPy and gradient descent. During training, the model updates its parameters to minimize the loss function, while the training performance is monitored using accuracy and loss curves. Finally, the system displays the original image, denoised image, convolution output, and training results, demonstrating the effectiveness of the proposed chest X-ray image enhancement and classification framework.

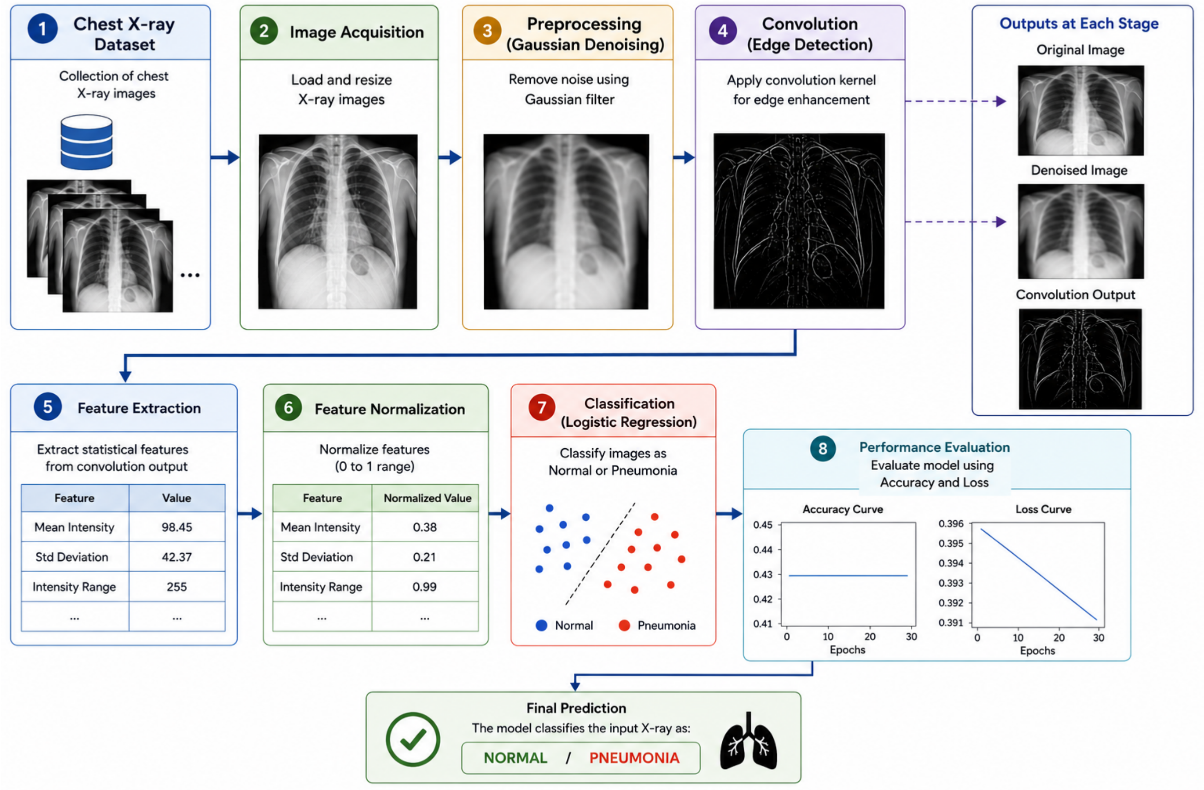

Figure 1: Block Diagram

4.3 Observations

The proposed system successfully enhanced chest X-ray images through Gaussian filtering and convolution-based edge detection. Gaussian filtering effectively reduced image noise while preserving important anatomical structures, resulting in clearer images for further processing. The convolution operation enhanced the edges and highlighted significant lung features, making them more suitable for feature extraction. The optimized OpenCV convolution method demonstrated better computational efficiency than the manually implemented convolution approach by reducing execution time.

Statistical features extracted from the processed images, including mean intensity, standard deviation, and intensity range, provided an effective representation for image classification. Feature normalization improved the stability of the logistic regression model during training. The loss curve showed a gradual decrease, indicating that the model was learning from the extracted features, while the accuracy remained nearly constant because sample labels were used instead of a real annotated medical dataset. Overall, the proposed workflow successfully integrated image enhancement, feature extraction, and machine learning concepts, demonstrating the feasibility of automated chest X-ray image analysis using Python.

5 Results and Discussion

The proposed chest X-ray image enhancement and classification system was successfully implemented in Python using OpenCV, NumPy, and Matplotlib. The preprocessing stage effectively removed noise from the chest X-ray images using Gaussian filtering, resulting in smoother images while preserving important anatomical structures. The convolution operation further enhanced the edges of the lungs and other significant regions, making the extracted features more informative for classification. A comparison between the manually implemented convolution and the optimized OpenCV convolution showed that the optimized method required significantly less execution time, demonstrating its computational efficiency.

The extracted statistical features, namely mean intensity, standard deviation, and intensity range, were successfully normalized and used to train the logistic regression classifier. The training process showed a gradual decrease in the loss curve, indicating that the model was learning from the input features. The accuracy remained nearly constant because the current implementation used sample (dummy) labels instead of a real annotated chest X-ray dataset. Nevertheless, the system successfully demonstrated the complete workflow of image enhancement, feature extraction, classification, and performance evaluation. The obtained results validate the feasibility of the proposed approach and provide a strong foundation for future work involving real medical datasets and advanced deep learning models such as Convolutional Neural Networks (CNNs).

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Program Outcomes and Relevance to the Complex Engineering Problem

PO No.	Program Outcome and Relevance to the Complex Engineering Problem
PO1	**Engineering Knowledge:** Applies knowledge of image processing, machine learning, and deep learning concepts to develop an automated chest X-ray image enhancement and classification system.
PO3	**Design/Development of Solutions:** Develops a Python-based framework that integrates Gaussian filtering, convolution, feature extraction, and logistic regression for chest X-ray image classification.
PO4	**Conduct Investigations of Complex Problems:** Evaluates the performance of image enhancement and classification methods using execution time, loss, and accuracy metrics.
PO5	**Modern Tool Usage:** Utilizes modern software tools such as Python, OpenCV, NumPy, and Matplotlib for implementing and visualizing the proposed system.
PO6	**Engineer and Society:** Demonstrates the application of artificial intelligence in healthcare to support medical image analysis and assist disease diagnosis.
PO11	**Project Management and Finance:** Plans and executes the project using open-source software tools, minimizing implementation cost while achieving project objectives.

6.2 Sustainable Development Goals (SDGs) Mapping

The proposed project focuses on enhancing and classifying chest X-ray images using image processing and machine learning techniques. Gaussian filtering and convolution are applied to improve image quality and extract important features. The extracted features are classified using a logistic regression model, and the system performance is evaluated using accuracy and loss metrics.

Table 2: Sustainable Development Goals (SDGs) and their Relevance

SDG No.	Sustainable Development Goal and Relevance
SDG 3	**Good Health and Well-being:** Enhances chest X-ray images and supports automated disease classification, enabling early diagnosis and improving healthcare services through artificial intelligence.
SDG 9	**Industry, Innovation and Infrastructure:** Promotes the use of image processing, machine learning, and deep learning techniques to develop innovative healthcare solutions.
SDG 4	**Quality Education:** Provides practical knowledge of medical image processing, artificial intelligence, and machine learning for research and academic learning.
SDG 10	**Reduced Inequalities:** Supports AI-assisted diagnosis that can improve access to healthcare services in rural and underserved regions.
SDG 17	**Partnerships for the Goals:** Encourages collaboration between healthcare professionals, researchers, and technology developers to build intelligent medical imaging systems.

7 Conclusion

The proposed Chest X-ray Image Enhancement and Classification Using Convolution, Machine Learning, and Deep Learning in Python was successfully implemented and evaluated. The system effectively enhanced chest X-ray images using Gaussian filtering for noise removal and convolution for edge detection, improving the visibility of important anatomical features. Statistical features extracted from the processed images were normalized and used to train a logistic regression classifier implemented using NumPy. The training results demonstrated a gradual reduction in the loss curve, indicating effective learning, while the comparison of manual and optimized convolution methods showed that the OpenCV implementation provided significantly faster execution.

Overall, the project successfully integrated digital image processing, feature extraction, and machine learning into a single framework for automated chest X-ray image analysis. The developed system demonstrates the practical application of artificial intelligence in medical imaging and provides a strong foundation for future enhancements. The project can be extended by using real annotated chest X-ray datasets and advanced deep learning models such as Convolutional Neural Networks (CNNs) to improve classification accuracy and support more reliable disease diagnosis in clinical environments.

Bibliography

[1] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, pp. 1097–1105, 2012.

[3] P. Rajpurkar, J. Irvin, K. Zhu *et al.*, "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," *arXiv preprint arXiv:1711.05225*, 2017.

[4] S. Candemir and S. Antani, "A Review on Lung Boundary Detection in Chest X-rays," *International Journal of Computer Assisted Radiology and Surgery*, vol. 14, no. 4, pp. 563–576, 2019.

[5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.